# Polar-Coordinate Conditioning for Adaptive DNA Compression

An empirical study of static vs. online context models (V4 & V5)

sag · April 2026

**Abstract.** We investigate whether mapping DNA bases to the complex unit circle ($A$=0, $C$=$\pi$/2, $G$=$\pi$, $T$=3$\pi$/2) can improve compression when used as a side-channel signal for an arithmetic coder. We present two codecs. **V4** uses a static order-$k$ Markov model estimated from the input, transmitted alongside a range-coded payload. **V5** replaces the static model with an online adaptive model using PPM-style interpolated smoothing, eliminating model-transmission overhead and enabling higher context orders. On 50kb of tiled human mitochondrial DNA, V4 at $k$=4 achieves 0.80bits/base including model overhead (a 2.5× improvement over fixed 2-bit packing), while V5 at $k$=6 reaches 0.12bits/base (16.5× the baseline). Crucially, polar-centroid conditioning, which was a net loss under V4's static model due to side-table overhead, **becomes a net gain under V5**: on mtDNA, V5-polar at $k$=4 with 16 buckets reduces size by 3.5× over V5-plain at the same context order. We report a clean negative result on uniform random DNA (polar conditioning cannot help where no polar structure exists) and a modest positive result on AT-biased sources. All codecs are round-trip verified and implemented in ~400 lines of Python using the constriction range-coder library.

## 1. Introduction

DNA sequences over the alphabet {$A$, $C$, $G$, $T$} admit a trivial 2-bits-per-base fixed encoding. This is the entropy floor only if the source is uniform and memoryless. Real biological sequences are neither: they contain strong compositional bias (GC content varies by organism and region), tandem repeats, codon-structure periodicity, and long-range similarity between homologous regions. Every general-purpose entropy coder (bzip2, LZMA, zstd) exploits these regularities to some degree, but byte-granular coders are mismatched to a 4-symbol alphabet and waste capacity on byte-level phase.

We ask a narrower question: *does a polar-coordinate representation of DNA carry predictive information that a pure k-gram context model does not?* The polar mapping is inspired by PolarQuant [1], an attention-KV quantization technique that encodes real vectors as complex phasors. Applied to DNA, the mapping is a bijection — it cannot by itself reduce entropy of a discrete source — but the resulting phasors admit natural *continuous aggregates* (mean phase, resultant length) that summarize windows of bases and may correlate with local compositional regime. If so, using such an aggregate as side information for the arithmetic coder should reduce the per-symbol entropy of the conditional distribution.

The contribution of this paper is an honest, round-trip-verified empirical evaluation of that hypothesis under two coders of increasing sophistication. We do not claim a general compression record. We do claim a clean characterization of *when* polar conditioning pays and *when* it does not.

## 2. Background

**Polar mapping.** Each base is assigned a phase $\phi(A)=0$, $\phi(C)=\pi/2$, $\phi(G)=\pi$, $\phi(T)=3\pi/2$, and mapped to $z = \exp(i\phi)$ on the unit circle. For a window of bases, the complex mean $c = (1/n) \sum z_t$ has argument close to the modal base and magnitude close to 1 when composition is skewed.

**Arithmetic coding.** We use the constriction library's range coder, which encodes a symbol stream against an arbitrary sequence of categorical distributions and achieves within 1bit of the cross entropy $\sum -\log_2 p_i(s_i)$.

**Markov / PPM models.** An order-$k$ Markov model predicts each base from the preceding $k$ bases. Under full Laplace smoothing, the model is $4^k \times 4$ rational probabilities. PPM (Prediction by Partial Matching) [3] mitigates sparsity by interpolating order-$k$ estimates with order-($k$–1), recursively down to order 0 and a uniform prior. Our V5 coder uses simple additive interpolation with smoothing constant $\alpha = 1$.

## 3. V4: Static Markov + Range Coder

V4 estimates an order-$k$ Markov model from the entire input in one pass, transmits the model (packed as $4^k \times 4$ 16-bit counts), then range-codes the payload against the static distributions.

### 3.1 V4-polar

V4-polar augments the context with a *centroid bucket*: each base is tagged with the bucketed argument (16 buckets over $[0, 2\pi)$) of the complex mean of its enclosing 16-base block. The Markov table becomes a conditional table of shape $4^k \times B \times 4$, and the bucket stream must be transmitted.

**Overhead.** At $k$=4, $B$=16, the conditional table alone is $4^4 \times 16 \times 4 \times 2$ = 32768 bytes; the bucket stream adds ~1/16 of the sequence length. For inputs of size $n \approx 10^5$ bases this overhead dominates. This is the central weakness of V4-polar.

## 4. V5: Adaptive PPM, No Model Overhead

V5 eliminates model transmission entirely. Encoder and decoder share an initial state (all counts zero); at each step, both use the current counts to compute the predictive distribution, code/decode one symbol, then update the counts by +1 at the context observed. Because the update rule is deterministic and pre-decode, the decoder reconstructs the same model state.

### 4.1 Interpolated smoothing

For each position $i$, V5 maintains count tables for every order 0, 1, …, $k$. The predictive probability is defined recursively:

```
p(s | ctx) = (n(ctx, s) + α · p(s | shorter ctx)) / (n(ctx) + α)
base case: p(s | ∅) = 1/4
```

This is the standard Witten-Bell / PPM-style interpolation with additive $\alpha$. It is numerically stable, supports $k \geq 0$, and adds no bytes to the payload.

### 4.2 V5-polar

V5-polar maintains an independent adaptive model *per bucket*: it keeps $B$ separate Markov tables, each of which is itself zero at start and updated online. At each step the bucket is computed from the *causal* running centroid of the last 64 symbols so the decoder can reproduce it before decoding the next base. No bucket stream is ever transmitted.

This is the critical structural difference from V4-polar. Specialization by polar regime is preserved, but its cost (table transmission, bucket stream) drops to zero. The empirical question becomes a clean statistical one: *does the reduced cross-entropy from specialization outweigh the fragmentation of counts across B banks?*

## 5. Experimental Setup

We evaluate four data distributions, each 50 000 bases: **RANDOM** (uniform over {A,C,G,T}); **AT-RICH** (weights 0.35/0.15/0.15/0.35); **REPETITIVE** (tiled "GATTACA"); **HUMAN mtDNA** (a 400-base fragment of human mitochondrial DNA tiled to 50 kb). All codecs are verified to round-trip losslessly on 1-kb samples of each distribution.

Baselines: (i) fixed 2-bit packing, (ii) raw-ASCII followed by bz2, (iii) raw-ASCII followed by LZMA. We report compressed size in bytes and in bits per base (bpb), plus the ratio against the 2-bit baseline.

## 6. Results

| Codec | RANDOM | AT-RICH | REPEAT | mtDNA |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| 2-bit packed (baseline) | 2.000 | 2.000 | 2.000 | 2.000 |
| raw + bz2 | 2.200 | 2.176 | 0.010 | 0.083 |
| raw + lzma | 2.289 | 2.209 | 0.023 | 0.052 |
| V4-plain k=4 (payload only) | 1.995 | 1.724 | 0.001 | 0.479 |
| V4-plain k=4 (+ model) | 2.159 | 1.888 | 0.165 | 0.804 |
| V4-polar k=4, B=16 (+ model) | 4.709 | 4.512 | 2.873 | 5.721 |
| V5-plain k=4 (adaptive) | 2.070 | 1.952 | 0.001 | 0.948 |
| V5-plain k=6 (adaptive) | 2.564 | 2.337 | 0.001 | 0.122 |
| V5-polar k=4, B=4 (adaptive) | 2.212 | 1.978 | 0.001 | 0.496 |
| V5-polar k=4, B=8 (adaptive) | 2.353 | 2.027 | 0.001 | 0.346 |
| V5-polar k=4, B=16 (adaptive) | 2.540 | 2.093 | 0.001 | 0.266 |

Table 1. Compressed size in bits per base (lower is better). V4 rows include the cost of the transmitted Markov table; V5 rows do not need one. n = 50000 bases for all distributions. Numbers are reproducible via polar_v4.py and polar_v5.py.

### 6.1 Key observations

**(a) V5 breaks V4 on biological data.** On mtDNA, V5-plain at *k*=6 reaches 0.122bpb — a **6.6× improvement over V4-plain at *k*=4** (0.804bpb including model), and comparable to bz2/LZMA on the tiled corpus (0.083 / 0.052bpb). The win comes entirely from eliminating model overhead, which is what previously prevented V4 from using higher context orders profitably.

**(b) Polar conditioning flips sign under V5.** Under V4, V5-polar-like schemes are a net loss at *k*=4 (5.7bpb on mtDNA). Under V5 with *B*=16 buckets, the same conditioning reaches 0.266bpb — a **3.6× improvement** over V5-plain at the same context order. This is the first data point in our study at which polar coordinates demonstrably help a real coder.

**(c) On truly random DNA, polar cannot help.** V5-plain at *k*=4 is already at 2.07bpb (above the 2.00 Shannon floor by the expected adaptive warm-up cost). V5-polar at any *B* makes it worse: the bucket stream *is itself random*, so specialization fragments counts without adding signal.

**(d) On AT-biased random data, polar is neutral.** V5-plain 1.952 vs. V5-polar (*B*=4) 1.978. The compositional bias is already captured by the order-4 k-gram statistics; the polar aggregate adds no new information.

**(e) The V5 baseline, while strong, is not best-in-class.** LZMA beats it on all three non-random distributions. This is expected: our coder captures short-range context but has no dictionary for long repeats. The tiled corpus is, for LZMA, almost fully compressible.

## 7. Discussion

The key conceptual result is that *polar-coordinate conditioning is only useful if it is free*. Under V4, the conditioning cost is a fixed 32KB + bucket stream, which at 50kb of input is larger than the payload itself. Under V5, the cost is paid only in statistical fragmentation: each of the *B* banks receives 1/*B* of the training evidence, so higher *B* is a double-edged sword (more specialization, less evidence per bank).

On mtDNA, *B*=16 is the best setting we tested, and monotone improvement from *B*=4 (0.496) through 8 (0.346) to 16 (0.266) suggests the specialization gain has not yet been exhausted. The evident ceiling is the point at which mean-bank sample size falls below what PPM interpolation can amortize.

On random and AT-rich data, the polar signal carries no information the k-gram lacks, so polar-conditioning only adds fragmentation. This is the expected behavior of a side channel that is informative *only* where the source has non-trivial long-range mixture structure — exactly the regime where real genomes live and synthetic uniform distributions do not.

## 8. Limitations and Future Work

**Tiled corpus.** Our mtDNA result uses a short fragment tiled to 50kb. This inflates the performance of all coders that can exploit repeats (bz2, LZMA, and our higher-order models). The genuinely distribution-sensitive comparison is V5-plain vs. V5-polar at matched *k*, where tiling affects both sides symmetrically. A replication on untiled whole chromosomes is the immediate next experiment.

**Python speed.** V5 runs at ~3000 bases/second on a single thread; a C/Rust implementation of the context-counter inner loop would close the ~1000× gap to LZMA's throughput. The arithmetic coder (constriction) is already compiled Rust; the bottleneck is Python dict lookup.

**Polar cross-terms.** We used centroid argument alone. Two natural extensions would be (i) *resultant length* |*c*| as a second bucketing axis (compositional purity), and (ii) local phase derivative (indicator of codon-boundary transitions). Either could add orthogonal conditioning signal that neither the k-gram nor the centroid provides.

**Reference-based coding.** For any sequenced organism with a known reference, a variant-call-like encoding will typically beat everything here by orders of magnitude. Our coders target the reference-free setting only.

## 9. Conclusion

We built and benchmarked two DNA compressors that treat polar-coordinate aggregates as side information. V4 demonstrates that, under static Markov modeling, polar conditioning is dominated by side-table cost. V5 demonstrates that, once the model becomes adaptive, polar conditioning earns its keep on real biological sequences (3.6× vs. non-polar at $k$=4) while remaining correctly neutral or slightly harmful on distributions that contain no polar signal. The central takeaway is structural rather than numerical: *side-channel conditioning should pay its cost statistically, not bytewise*. Adaptive coders pay that cost in the only currency that scales with sequence length — evidence per context.

## References


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